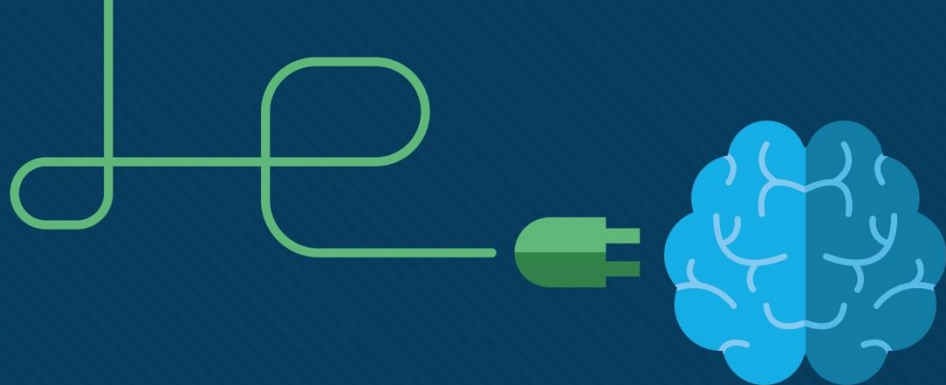




Switching Concepts



Frame Forwarding

Switching in Networking

Two terms are associated with frames entering or leaving an interface:

- **Ingress** – entering the interface
- **Egress** – exiting the interface

A switch forwards based on the ingress interface and the destination MAC address.

A switch uses its MAC address table to make forwarding decisions.

Note: A switch will never allow traffic to be forwarded out the interface it received the traffic.



Port Table

Destination Addresses	Port
EE	1
AA	2
BA	3
EA	4
AC	5
AB	6

The Switch MAC Address Table

A switch will use the destination MAC address to determine the egress interface.

Before a switch can make this decision it must learn what interface the destination is located.

A switch builds a MAC address table, also known as a Content Addressable Memory (CAM) table, by recording the source MAC address into the table along with the port it was received.

The Switch Learn and Forward Method

The switch uses a two step process:

Step 1. Learn – Examines Source Address

- Adds the source MAC if not in table
- Resets the time out setting back to 5 minutes if source is in the table

Step 2. Forward – Examines Destination Address

- If the destination MAC is in the MAC address table it is forwarded out the specified port.
- If a destination MAC is not in the table, it is flooded out all interfaces except the one it was received.

Table d'auto-apprentissage – CAM Table

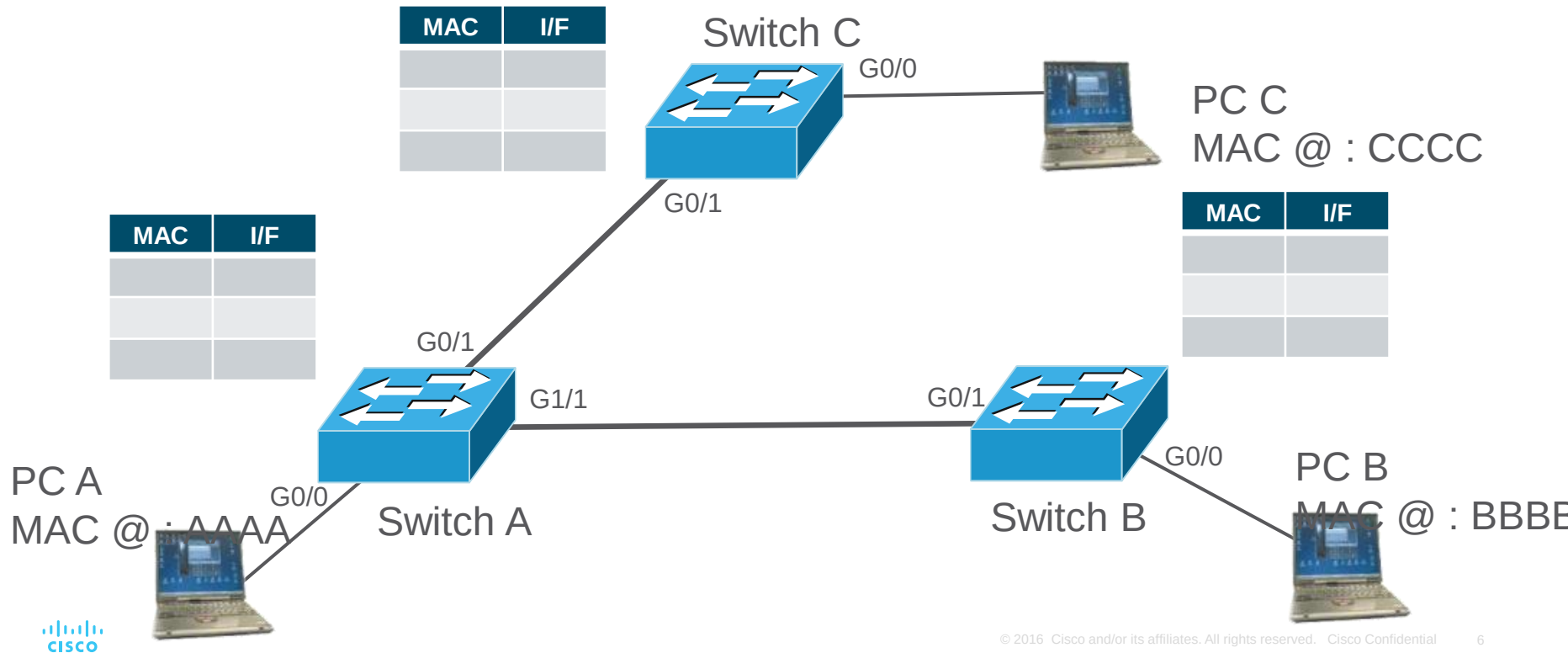


Table d'auto-apprentissage – CAM Table

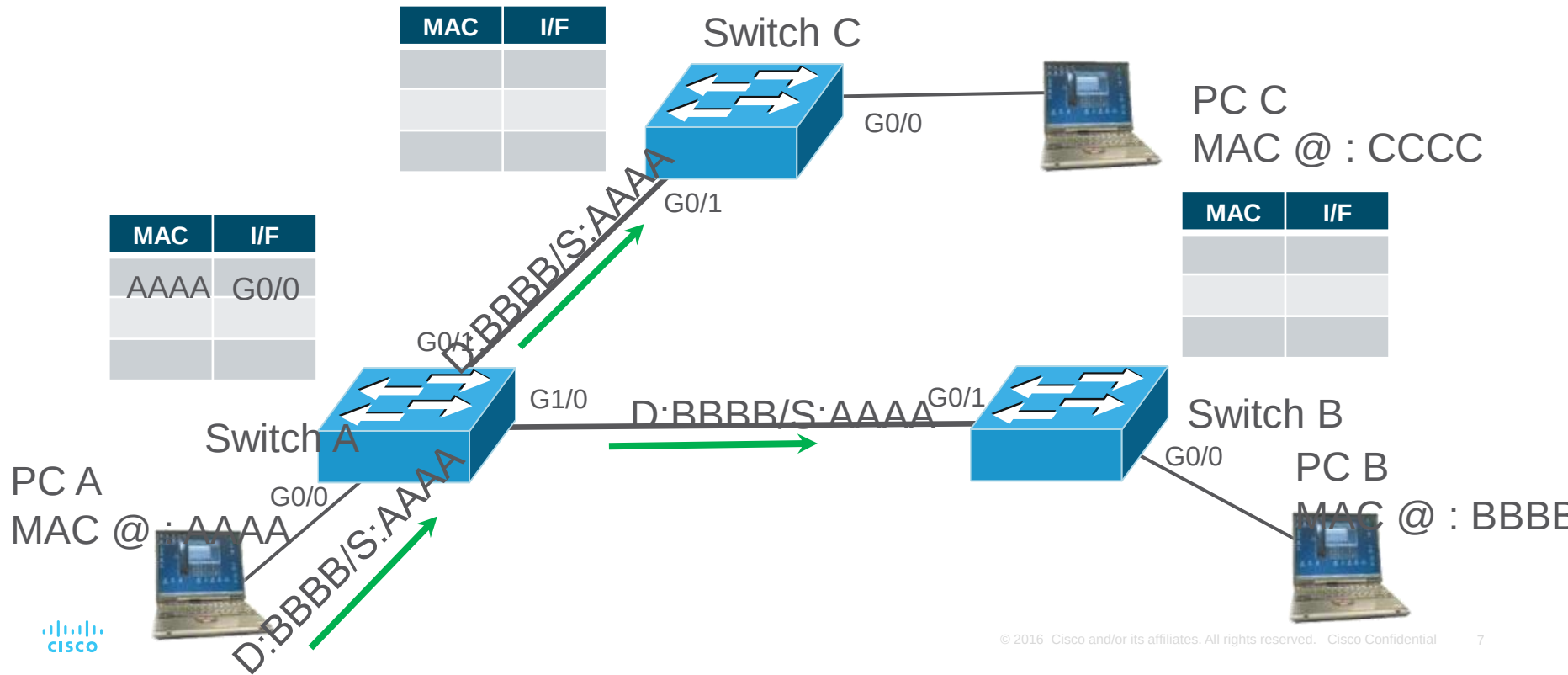


Table d'auto-apprentissage – CAM Table

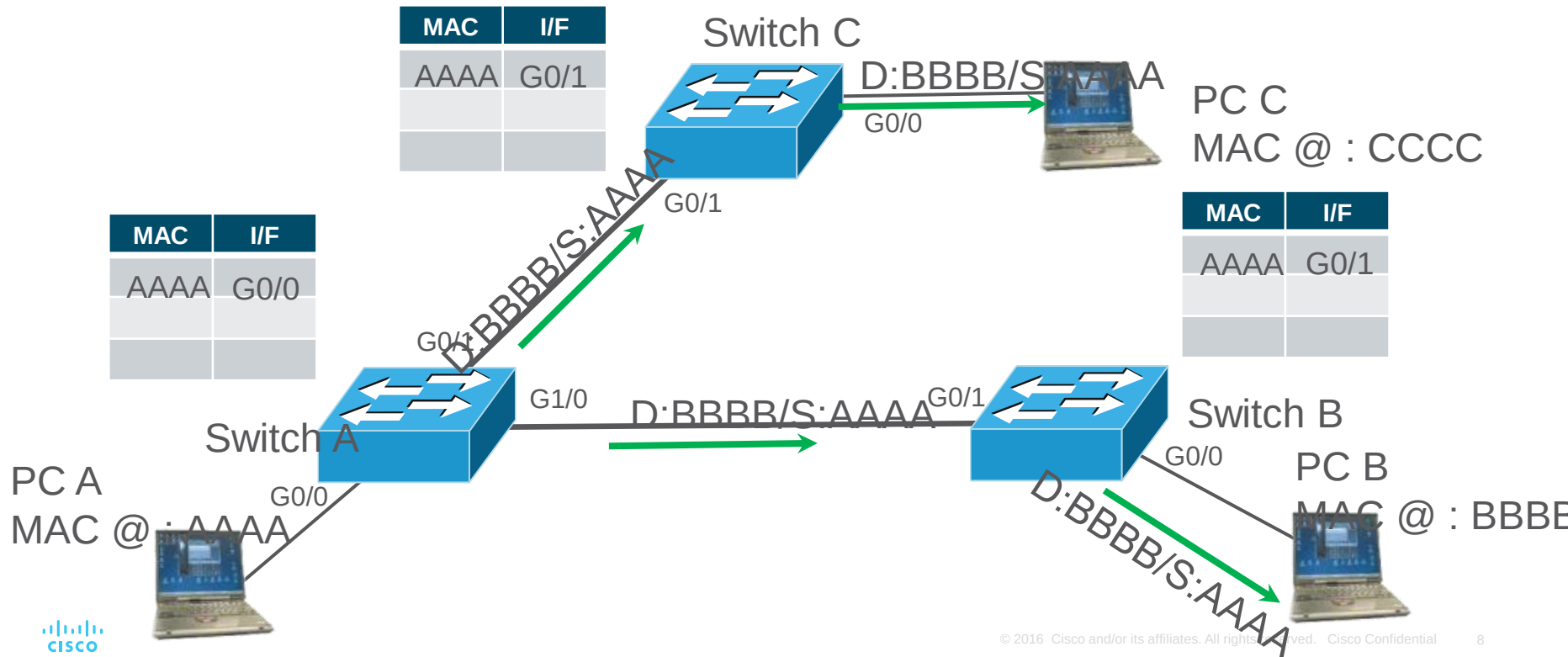


Table d'auto-apprentissage – CAM Table

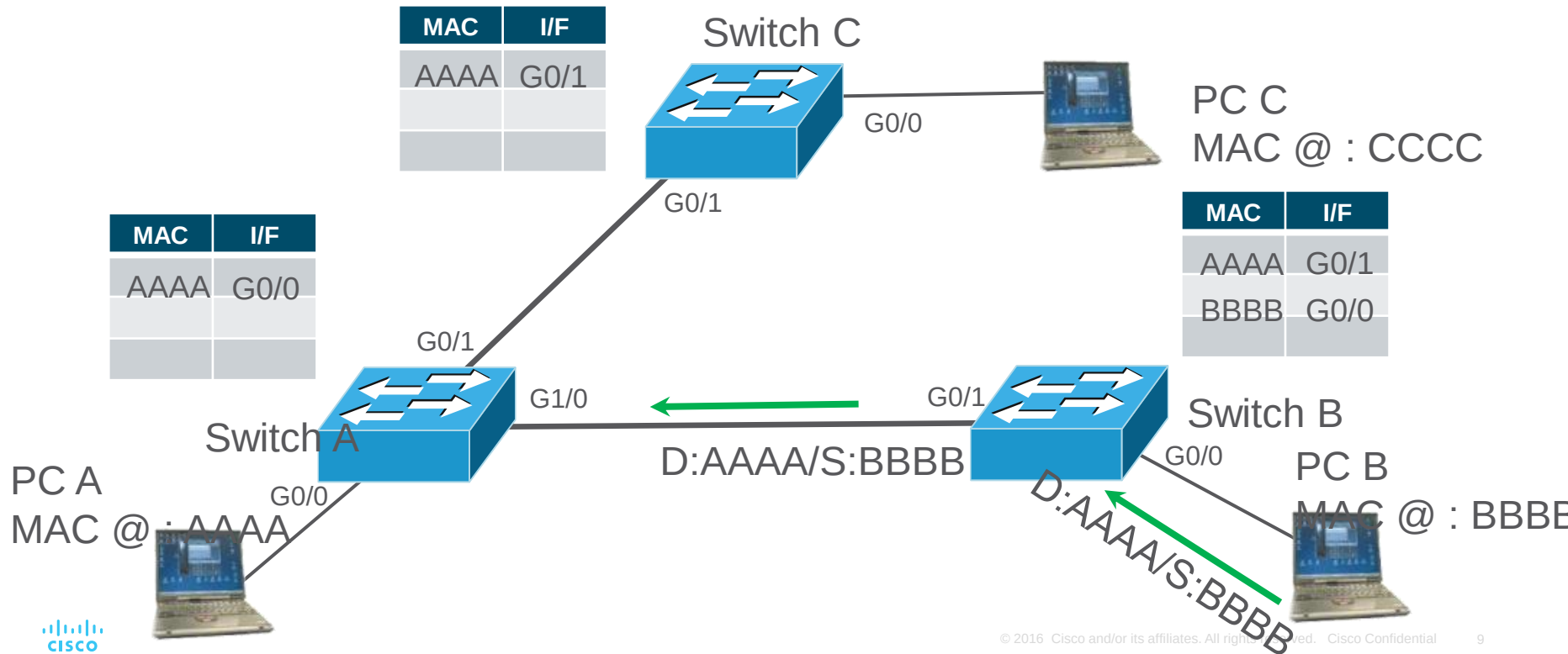
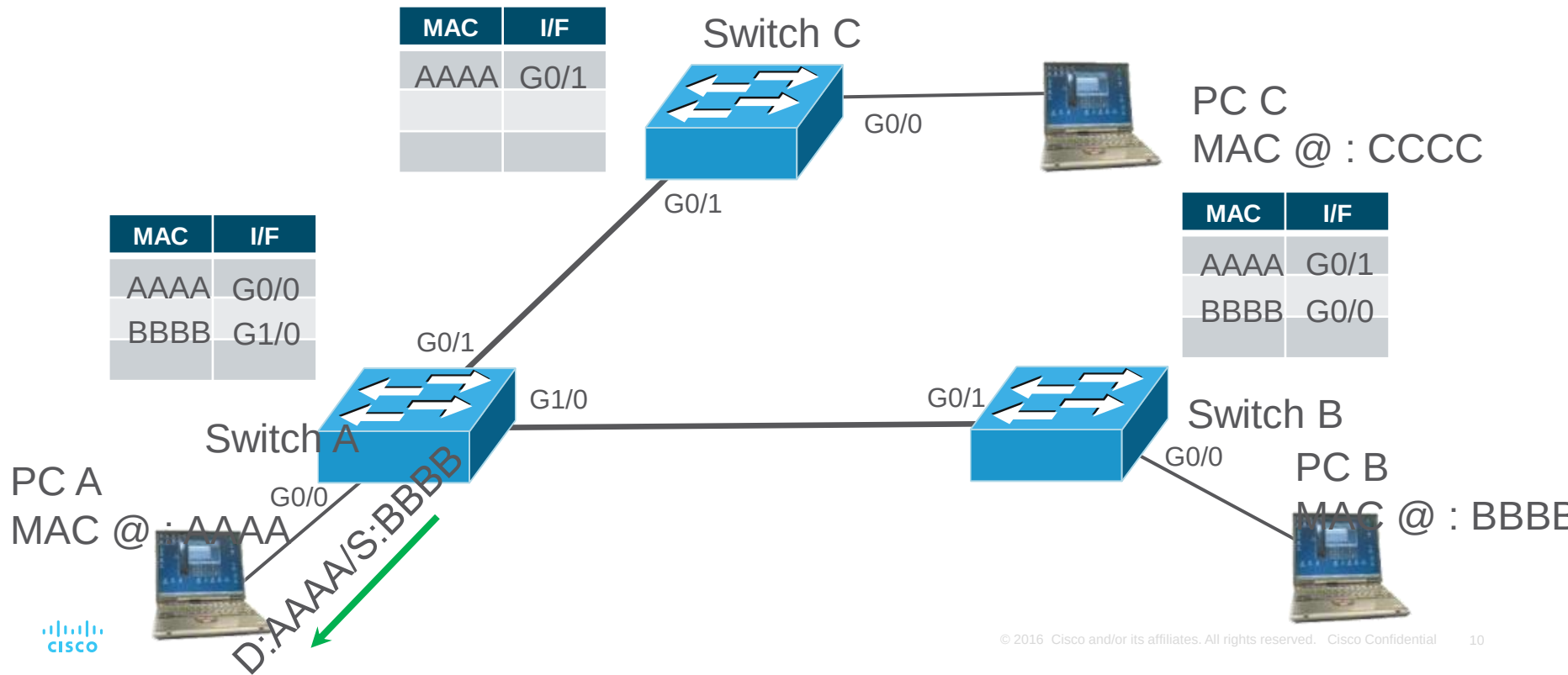


Table d'auto-apprentissage – CAM Table



Switch Forwarding Methods

Switches use software on application-specific-integrated circuits (ASICs) to make very quick decisions.

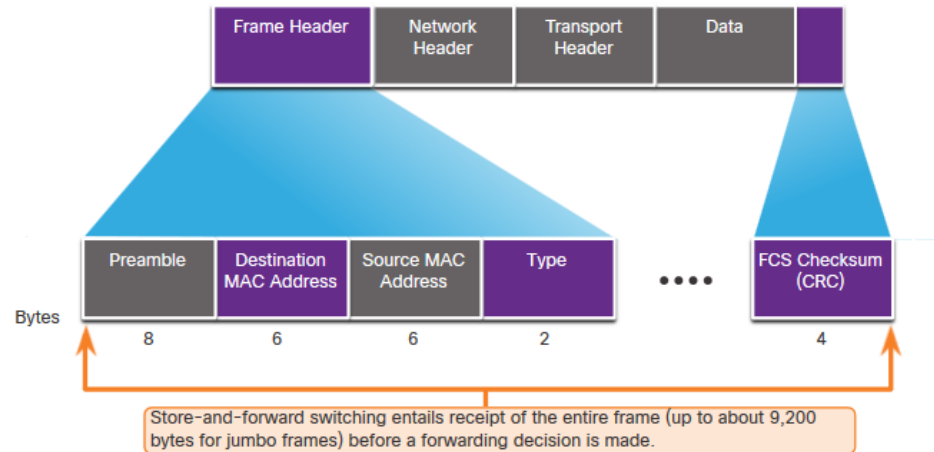
A switch will use one of two methods to make forwarding decisions after it receives a frame:

- **Store-and-forward switching** - Receives the entire frame and ensures the frame is valid. Store-and-forward switching is Cisco's preferred switching method.
- **Cut-through switching** – Forwards the frame immediately after determining the destination MAC address of an incoming frame and the egress port.

Store-and-Forward Switching

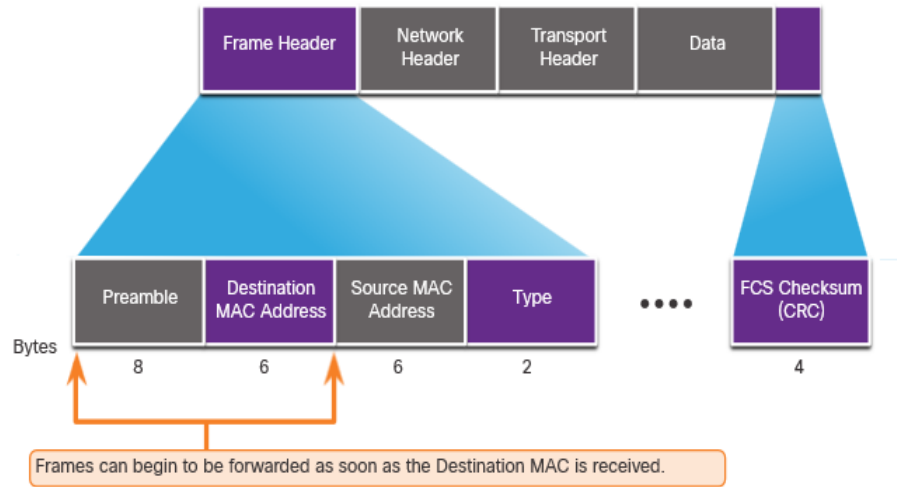
Store-and-forward has two primary characteristics:

- Error Checking – The switch will check the Frame Check Sequence (FCS) for CRC errors. Bad frames will be discarded.
- Buffering – The ingress interface will buffer the frame while it checks the FCS. This also allows the switch to adjust to a potential difference in speeds between the ingress and egress ports.



Frame Forwarding

Cut-Through Switching



- Cut-through forwards the frame immediately after determining the destination MAC.
- Fragment (Frag) Free method will check the destination and ensure that the frame is at least 64 Bytes. This will eliminate runs.

Concepts of Cut-Through switching:

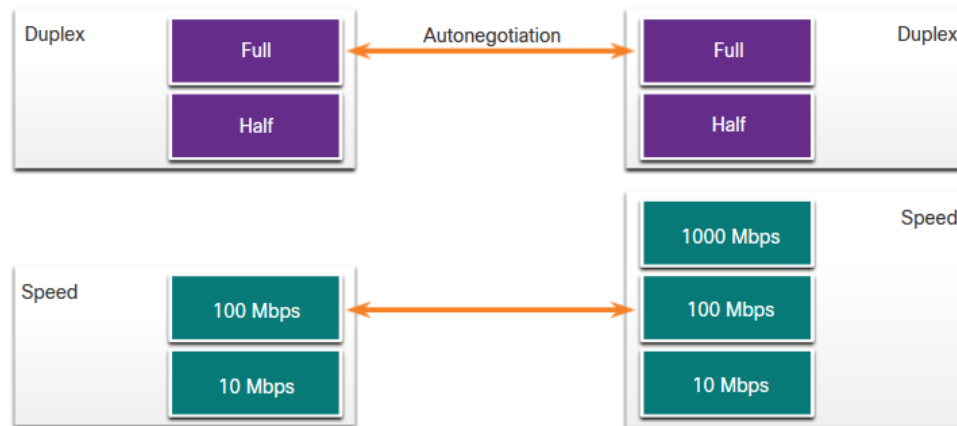
- Is appropriate for switches needing latency to be under 10 microseconds
- Does not check the FCS, so it can propagate errors
- May lead to bandwidth issues if the switch propagates too many errors
- Cannot support ports with differing speeds going from ingress to egress

Switching Domains

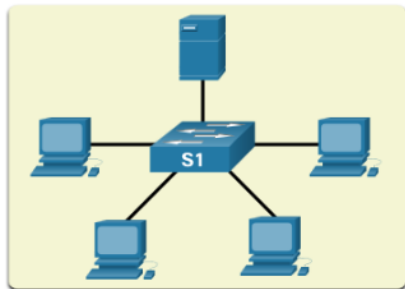
Collision Domains

Switches eliminate collision domains and reduce congestion.

- When there is full duplex on the link the collision domains are eliminated.
- When there is one or more devices in half-duplex there will now be a collision domain.
 - There will now be contention for the bandwidth.
 - Collisions are now possible.
- Most devices, including Cisco and Microsoft use auto-negotiation as the default setting for duplex and speed.



Broadcast Domains



- A broadcast domain extends across all Layer 1 or Layer 2 devices on a LAN.
- Only a layer 3 device (router) will break the broadcast domain, also called a MAC broadcast domain.
- The broadcast domain consists of all devices on the LAN that receive the broadcast traffic.
- When the layer 2 switch receives the broadcast it will flood it out all interfaces except for the ingress interface.
- Too many broadcasts may cause congestion and poor network performance.
- Increasing devices at Layer 1 or layer 2 will cause the broadcast domain to expand.

Alleviated Network Congestion

Switches use the MAC address table and full-duplex to eliminate collisions and avoid congestion.

Features of the switch that alleviate congestion are as follows:

Protocol	Function
Fast Port Speeds	Depending on the model, switches may have up to 100Gbps port speeds.
Fast Internal Switching	This uses fast internal bus or shared memory to improve performance.
Large Frame Buffers	This allows for temporary storage while processing large quantities of frames.
High Port Density	This provides many ports for devices to be connected to LAN with less cost. This also provides for more local traffic with less congestion.

LAN Architecture

Local Area Network

- Network within a limited area, from a few meters (home network) to a perimeter of 1 or 2 kms (Campus)
- Most of the time owned, and operated, by the company using it
- This company has full responsibility for the well-functioning :
 - Buying of equipments
 - Installing or monitoring of installation
 - Daily operation
 - Remediation if problems
- Contract with maintenance companies for service delivery (technical support)

Switches for LAN

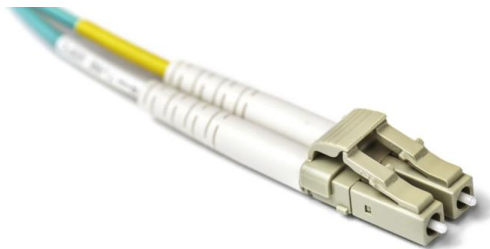
- The main hardware component for LAN is the switch
- It is made of a CPU, RAM memory, storage unit (SSD),...
- Its function is to connect end devices (PC, servers,..) and other switches, thanks to ports
- Depending the type of cable, the ports can be RJ45 (copper), ST/SC/LC (fiber optic),..
- Different form factors :
 - Fixed, with 12, 24, 48,.. ports
 - Modular, in chassis, with card in slots, each card with its type of ports

Connectors and cables

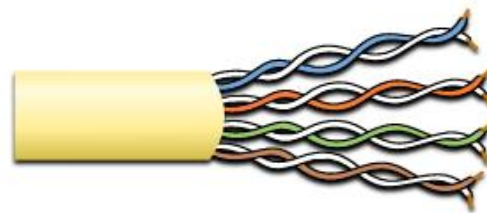
- RJ45 connectors for copper cables
- LC connectors for fiber optics



RJ45 connector



Fiber optic Connectors



Copper cable

Fixed configuration switch

- Example of a fixed configuration product
- 48 RJ45 ports to connect copper cables, usually to PC, printers,..
- 2 or 4 ST/SC/LC ports to connect fiber optics, usually to other switches

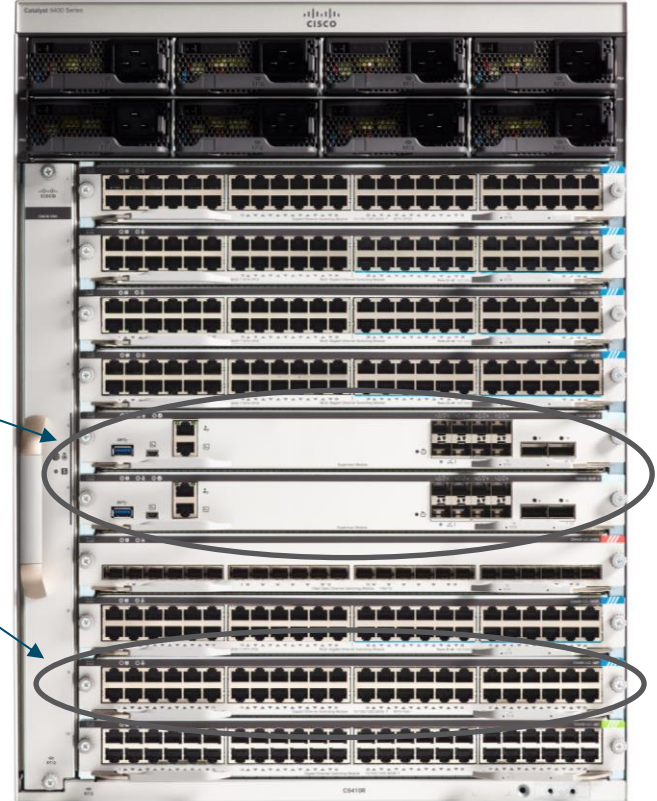
48 RJ45 ports

2 or 4 ST/SC/LC ports.
Possibility to accommodate
other connectivities



Modular configuration switch

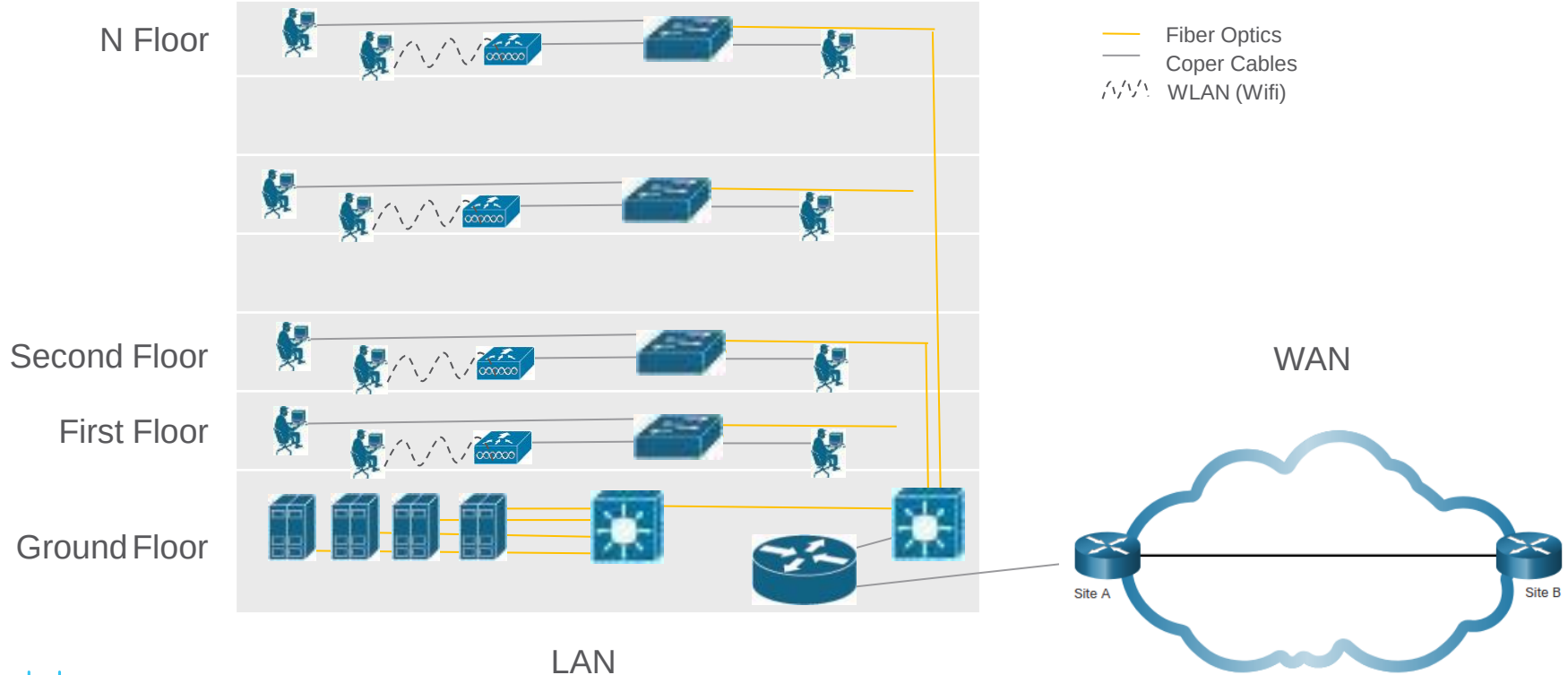
- Example of a modular product
- Made of an empty chassis, with modular cards
- Supervisor card with CPU, RAM,.... Can be duplicated for redundancy
- Line cards with connectivity ports
- Possibility to mix any kinds of cards (RJ45, fiber optics,..) to match required configuration



Choice of switch platform

- All platforms share the same Operating System, with same functionalities
- A network made of switches of different kinds will then guaranty full interoperability
- The differences in hardware will limit possibilities of a given product
- Recent evolutions of Ethernet standard offer speeds up to several 100 Gbps !!!!
- The form factor will also dictate where to use the product (see later)

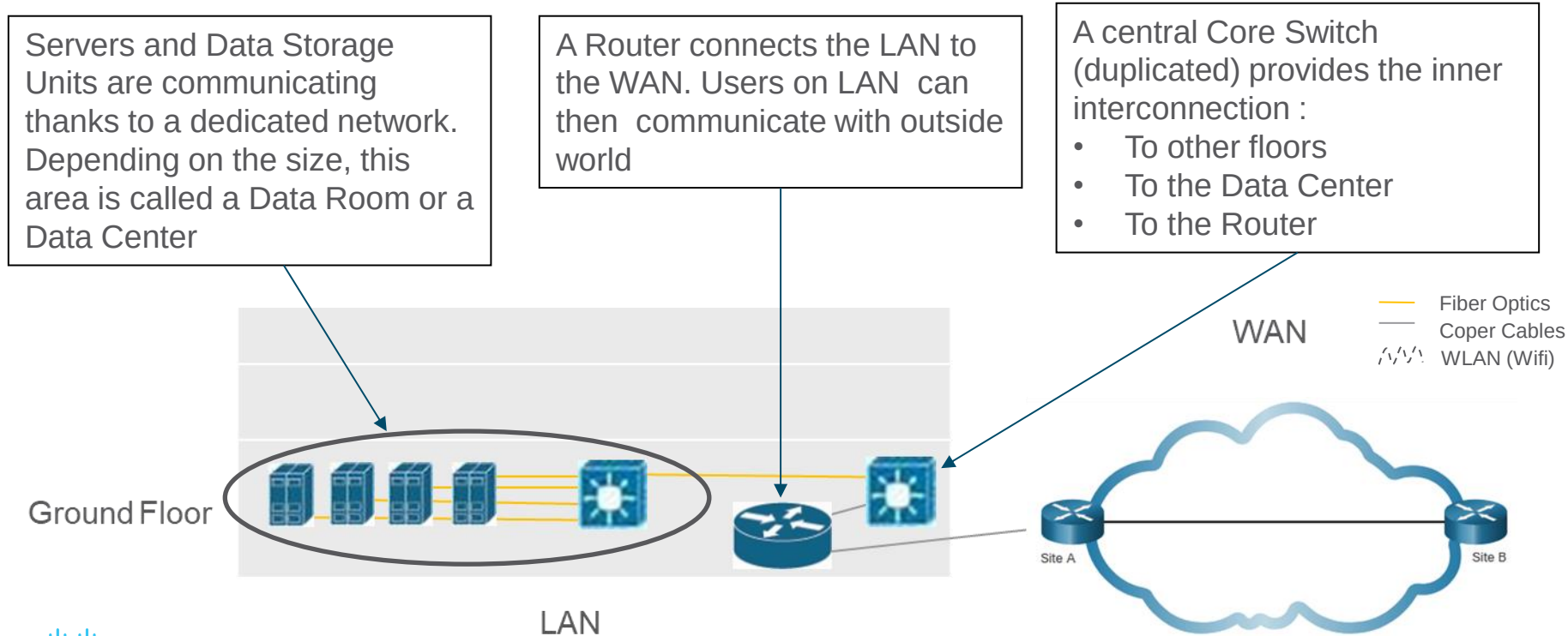
LAN in a building



Ground Floor infrastructure

- Sometimes is in the basement, for better physical protection
- Private room, with badges
- All critical IT resources :
 - Servers, processing data
 - Data storage units
 - Central switch (Core or Backbone)
- Telecom lines to communicate with outside world
 - WAN link for router
 - Phone lines

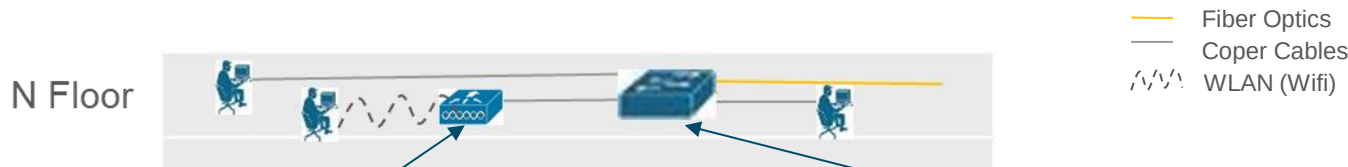
Ground Floor infrastructure



Other Floor infrastructure

- Where users are located
- Employees can use Personal Computers, Laptops, Smartphones, Tablets
- Most of the time, printers also available on same floor
- A Switch, called Access Switch, provides connectivity to users :
 - Copper cable (aka Ethernet cable) for PC and printers
 - WLAN (aka Wifi) for Laptops, Smartphones and Tablets (possibly printers)
 - Fiber optic connections between floors

Other Floor infrastructure



Access Point is the device interconnecting cable network (switch) with radio signal (Wifi)

The Access Switch is located in a dedicated room, preferably locked.

RJ45 ports connect copper cables to :

- Personal Computers
- Printers
- Access Points, providing WLAN (aka Wifi) connection

Fiber Optic ports connects to Core Switch, at ground floor

Key takeaway

- Cabling system is made of :
 - Copper cables on same floor
 - Fiber Optics between ground floor and other floors
- On same floor, cable lengths remains in limit of 100 meters (Ethernet standard)
- Between floors, distance and perturbations require use of fiber optics

Key takeaway

- WLAN (Wifi) is not a substitute for cable network, but an extension
- Core switch is duplicated, in case of a failure
- Switch architecture is made of two layers :
 - Access switch (users)
 - Core Switch (backbone)

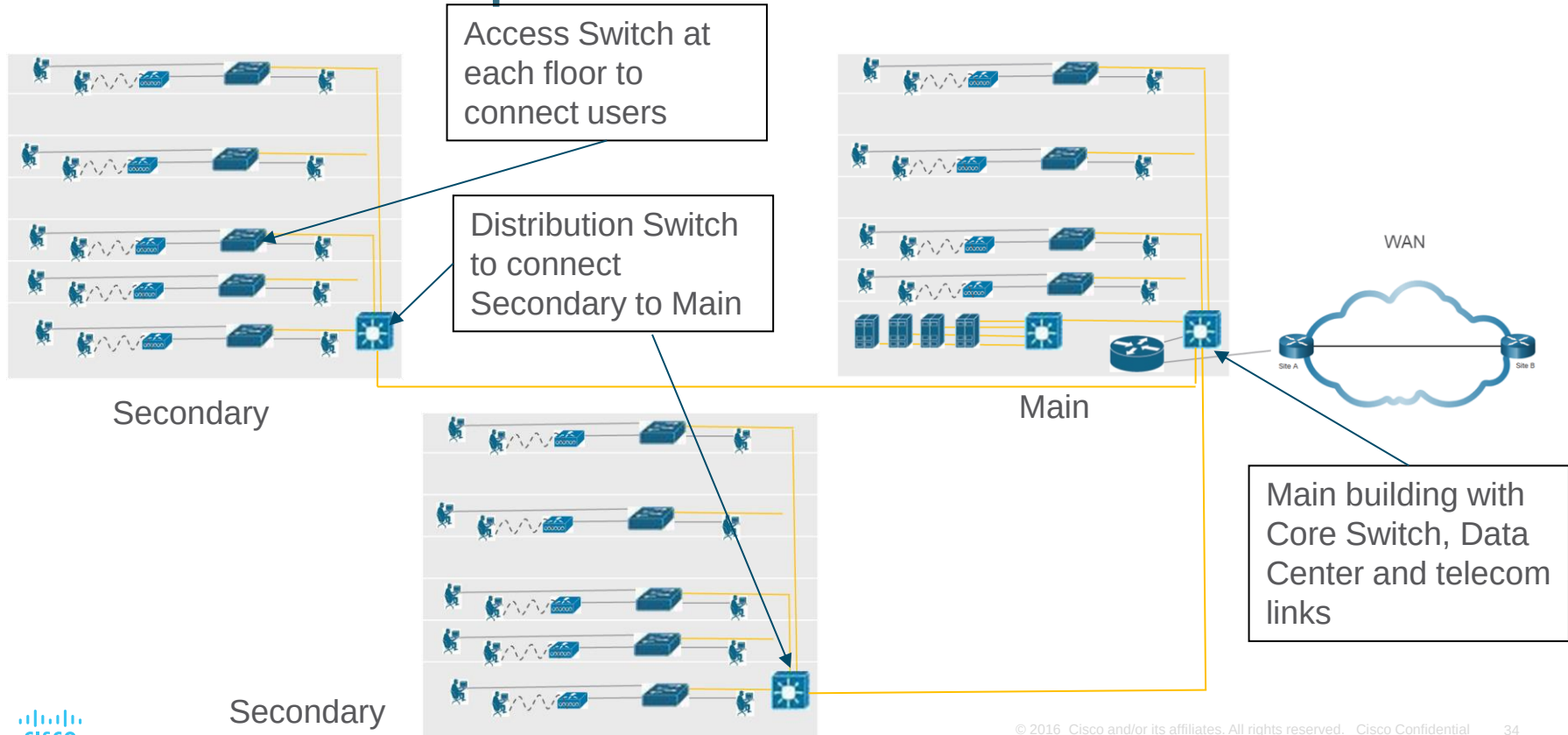
LAN in a Campus

- LAN design for a Campus is based upon same rules
 - One of the buildings is the main one, with telecom links, and federates other buildings
 - Ground floor of secondary buildings will be the boundary between floors and main

Campus map



LAN in a Campus



Key takeaway

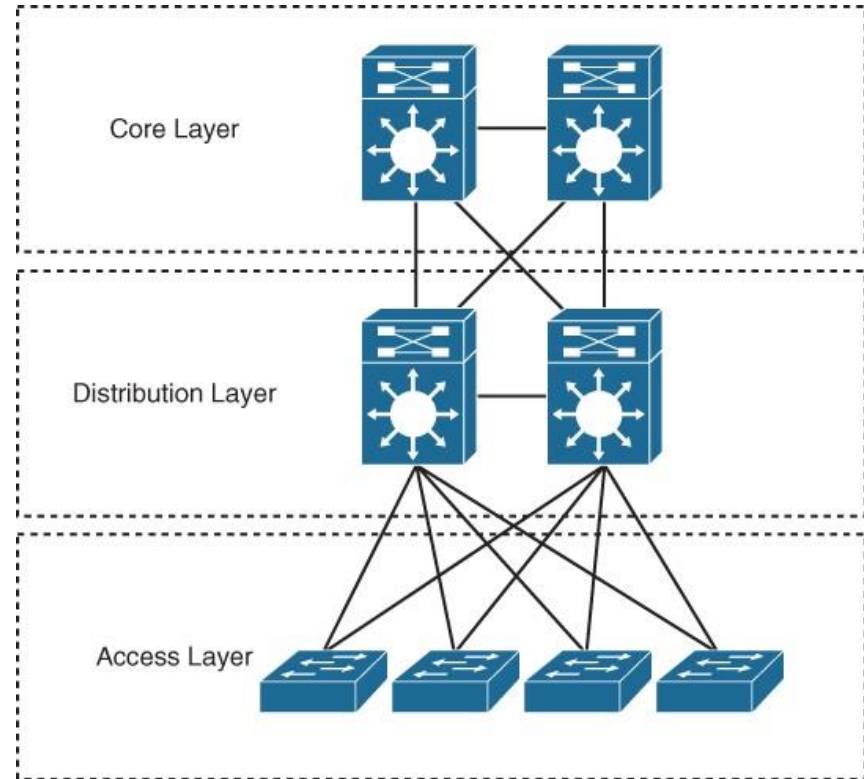
- Cabling system keeps the same logic
- Use of fiber optics between buildings (distance)
- Multi-mode fiber optics dictates the limit of 2 kms
- Main building with Data Center and telecom links
- Best practice is to have only users in Secondary buildings : no servers outside Data Center

Key takeaway

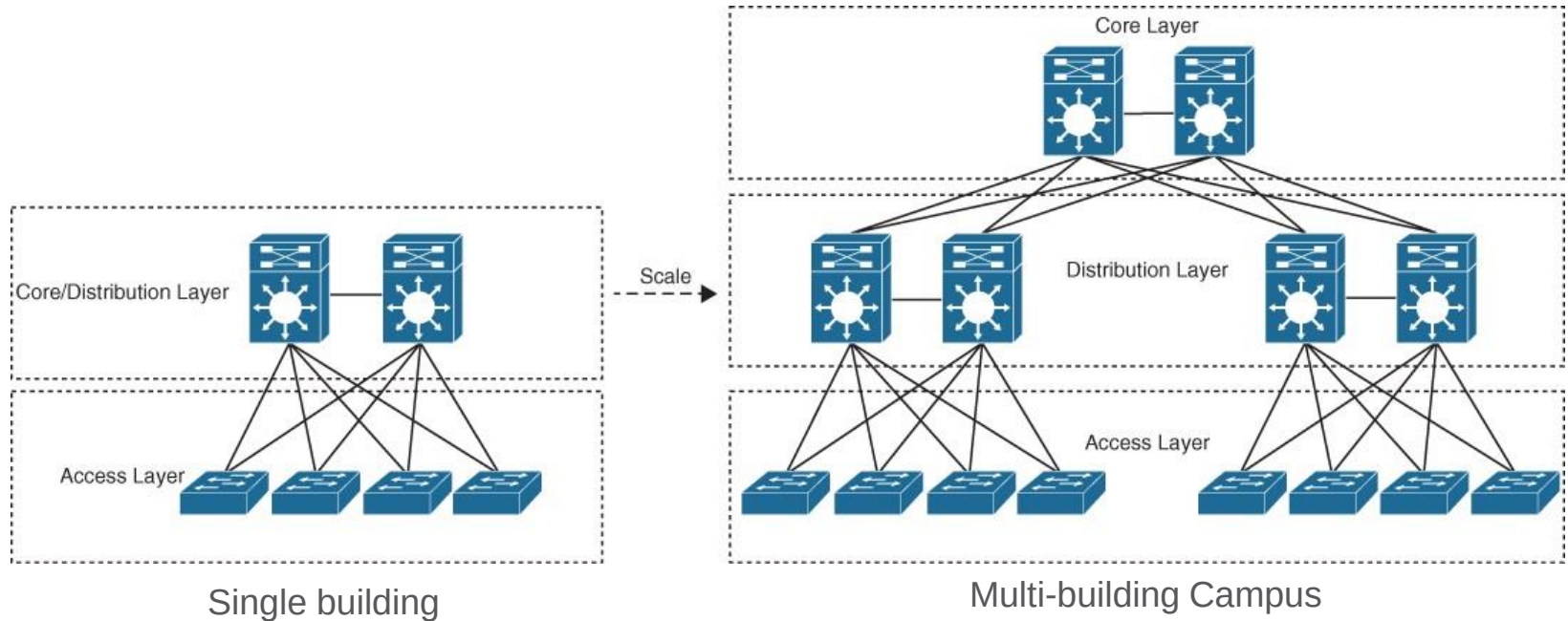
- Core and Distribution switches are duplicated
- Switch architecture made of three layers :
 - Access switch (users)
 - Distribution (agregation)
 - Core Switch (backbone)
- Architecture is fault tolerant and scalable

Hierarchical LAN Design Model

- **Access layer** - Gives endpoints and users direct access to the network.
- **Distribution layer** - Provides an aggregation point for the access layer and acts as a services and control boundary between the access layer and the core layer.
- **Core layer (also referred to as the backbone)** - Provides connections between distribution layers for large environments.

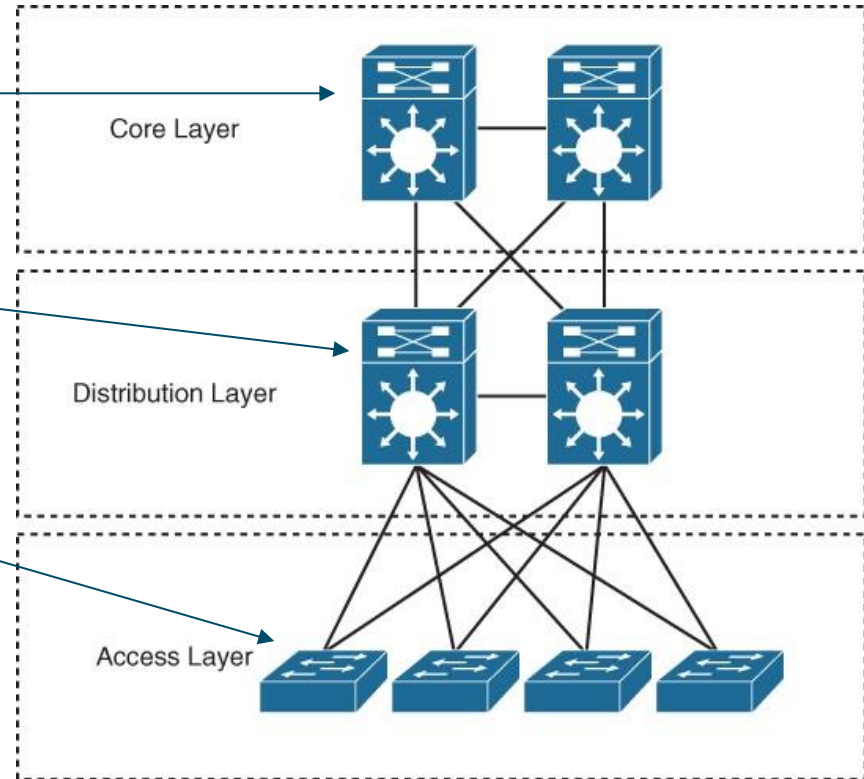


Scalable Layer Design



How to match switches with roles ?

- Core switches : high-end modular
- Distribution switches : mid-range modular, possibly high-end
- Access switches : fixed form factor, possibly mid-range modular

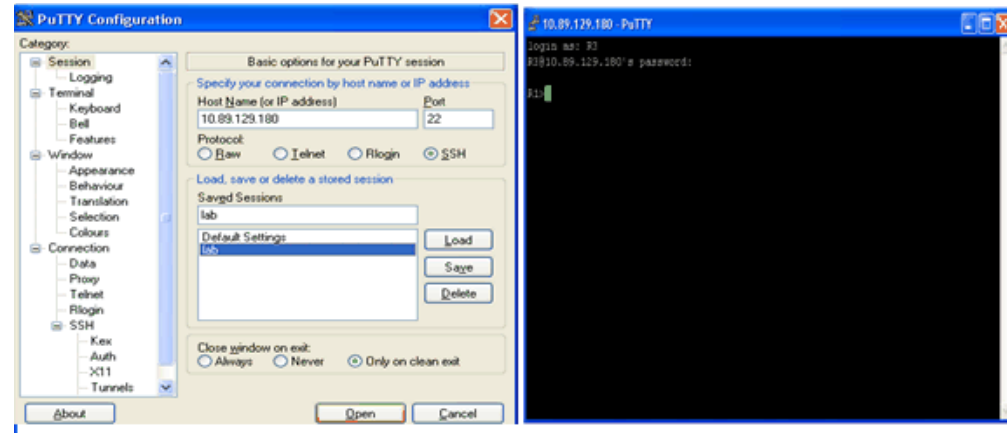


Basic Switch Configuration

Cisco IOS Access

Access Methods

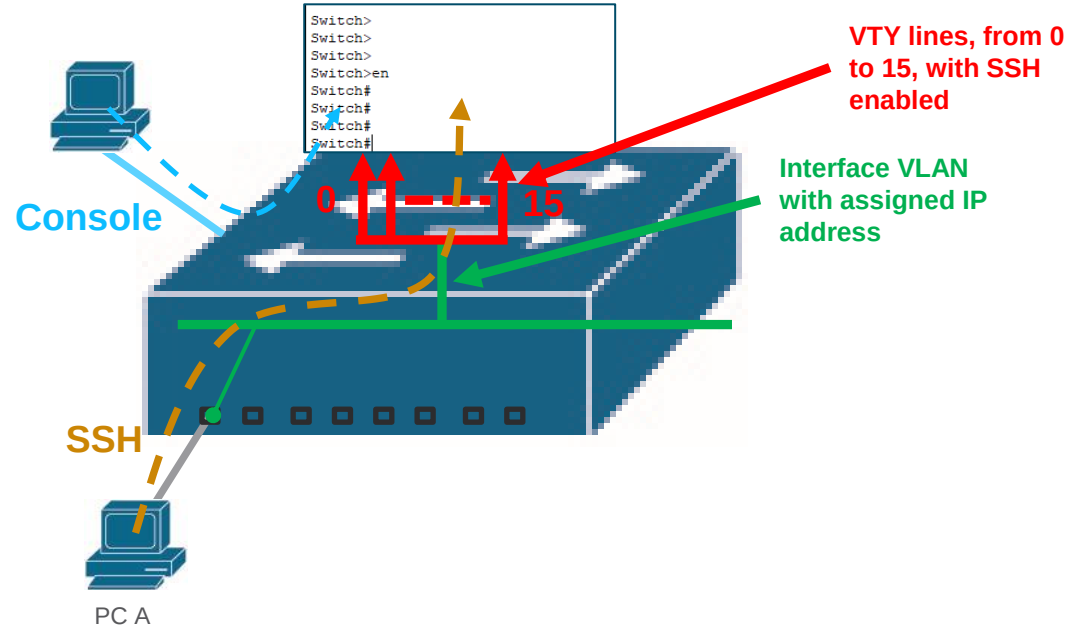
- **Console** – A physical management port used to access a device in order to provide maintenance, such as performing the initial configurations.
- **Secure Shell (SSH)** – Establishes a secure remote CLI connection to a device, through a virtual interface, over a network. (Note: This is the recommended method for remotely connecting to a device.)
- **Telnet** – Establishes an insecure remote CLI connection to a device over the network. (Note: User authentication, passwords and commands are sent over the network in plaintext.)



Secure Remote Access

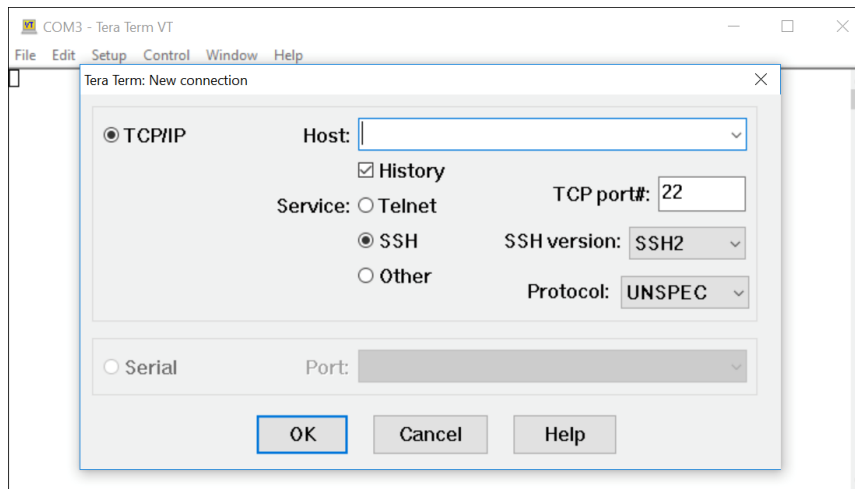
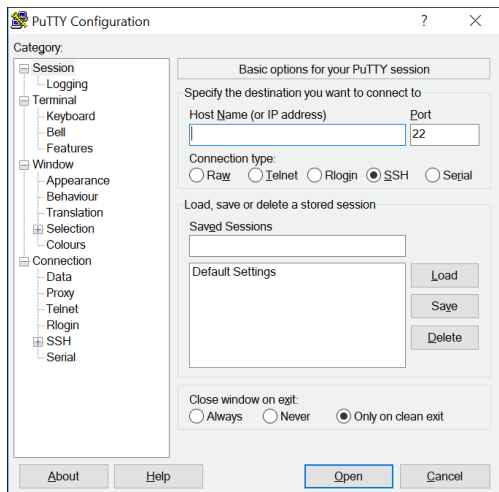
Access to the CLI

- Administrator initiates a SSH session with the switch
- The target IP address is Interface VLAN
- Then, VTY lines, with number from 0 to 15, challenge the connection with SSH and username/password
- On routers, only 5 VTY lines, numbered from 0 to 4+
- But first configuration must be done via console port



Terminal Emulation Programs

- Terminal emulation programs are used to connect to a network device by either a console port or by an SSH/Telnet connection.
- There are several terminal emulation programs to choose from such as PuTTY, Tera Term and SecureCRT.



Primary Command Modes

User EXEC Mode:

- Allows access to only a limited number of basic monitoring commands
- Identified by the CLI prompt that ends with the > symbol

```
Router>
```

```
Switch>
```

Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#
```

```
Switch#
```

Configuration Mode and Subconfiguration Modes

Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config) #
```

Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line) #
```

Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if) #
```

Navigation Between IOS Modes

▪ Privileged EXEC Mode:

- To move from user EXEC mode to privilege EXEC mode, use the **enable** command.

```
Switch> enable  
Switch#
```

▪ Global Configuration Mode:

- To move in and out of global configuration mode, use the **configure terminal** command. To return to privilege EXEC mode, use the **exit** command.

```
Switch(config)#  
Switch(config)#exit  
Switch#
```

▪ Line Configuration Mode:

- To move in and out of line configuration mode, use the **line** command followed by the management line type. To return to global configuration mode, use the **exit** command.

```
Switch(config)#line console 0  
Switch(config-line)#exit  
Switch(config)#
```

Navigation Between IOS Modes (Cont.)

Subconfiguration Modes:

- To move out of any subconfiguration mode to get back to global configuration mode, use the **exit** command. To return to privilege EXEC mode, use the **end** command or key combination **Ctrl +Z**.
- To move directly from one subconfiguration mode to another, type in the desired subconfiguration mode command. In the example, the command prompt changes from **(config-line)#** to **(config-if)#**.

```
Switch(config)#line console 0
Switch(config-line)#end
Switch#
```

```
Switch(config-line)#interface FastEthernet 0/1
Switch(config-if)#
```

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Configure Passwords (Cont.)

Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password password** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Basic Device Configuration

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd** *# the message of the day* # global config command.

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```

Save Configurations

Configuration Files

- There are two system files that store the device configuration:
 - **startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
 - **running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
 - To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

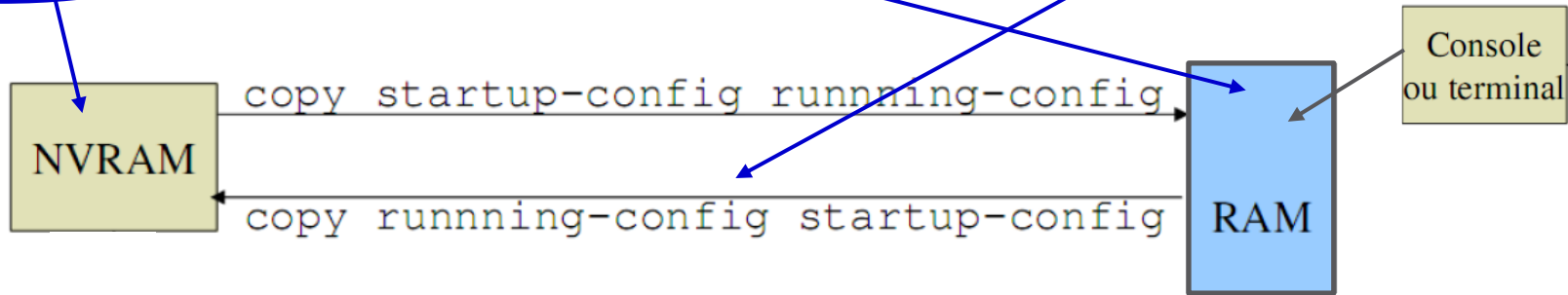
Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

Save Configurations

Configuration Files

- Location of system files :

- startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
- running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
- To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.



Switch Virtual Interface Configuration

To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask command*.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```

